

Lecture 2: Future Value, Present Value, and Interest Rates

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Overview

Interest rates are of enormous importance to virtually everyone:

- Link present to future - Allow us to compare payments made on different dates
- Future reward - Tell us the return for lending today
- Cost of borrowing - Show the cost of borrowing now and repaying later

Key concepts:

- Present value and future value
- Bond valuation
- Relationship between inflation and interest rates

The Road Ahead

1. Valuing Monetary Payments Now and in the Future
2. Applying Present Value
3. Real and Nominal Interest Rates

Future Value

Future value (FV) is the value on some future date of an investment made today.

Formula: For a present value (PV) invested at interest rate i :

$$FV = PV \times (1 + i)$$

Example: \$100 one year from now at 5% annual interest:

$$FV = \$100 \times (1.05) = \$105$$

The higher the interest rate or the amount invested, the higher the future value.

Compound Interest

Two-year example: \$100 at 5% for two years:

$$\begin{aligned} \$110.25 &= \underbrace{\$100}_{\text{PV}} + \underbrace{\$5}_{\text{1st year interest}} + \underbrace{\$5}_{\text{2nd year interest}} + \underbrace{\$0.25}_{\text{compound interest}} \\ &= \$100 \times (1.05)^2 \end{aligned}$$

Compound interest is the interest earned on interest.

General formula for n years:

$$FV_n = PV \times (1 + i)^n$$

Time Units and Basis Points

Caution: Interest rate and n must use the same time units.

Example: Monthly interest rate i^m for one year (12 months):

$$\$100 \times (1 + i^m)^{12} = \$100 \times (1.05)$$

Solving gives $i^m = 0.41\%$ or 41 basis points

Basis point: One-hundredth of a percentage point (0.01%)

Present Value

Present value (PV) is the value today of a payment promised in the future.

Formula: For a future value (FV) with interest rate i :

$$PV = \frac{FV}{1 + i}$$

Example: \$105 one year from now at 5%:

$$PV = \frac{\$105}{1.05} = \$100$$

Key insight: Present value falls as the interest rate rises.

Present Value: General Formula

For n years at annual interest rate i :

$$PV = \frac{FV_n}{(1 + i)^n}$$

Example: \$105 in two years at 5%:

$$PV = \frac{\$105}{(1.05)^2} = \$95.24$$

Present value is higher when:

- Future value is higher
- Time until payment is shorter
- Interest rate is lower

Internal Rate of Return

Internal rate of return equates the present value of an investment with its cost.

Example: \$225 loan with two repayment options:

1. \$100 per year for 3 years
2. \$125 per year for 2 years

Option 1: Find i such that:

$$\$225 = \frac{\$100}{1+i} + \frac{\$100}{(1+i)^2} + \frac{\$100}{(1+i)^3}$$

Solution: $i = 15.9\%$

Option 2: Find i such that:

$$\$225 = \frac{\$125}{1+i} + \frac{\$125}{(1+i)^2}$$

Solution: $i = 7.3\%$

Conclusion: Option 1 is more profitable (higher return).

Bonds

A bond is a promise to make a series of payments on specific future dates.

Coupon bond terminology:

- **Coupon payments** - Annual payments by bond issuer
- **Coupon rate** - Annual payments as % of amount borrowed (e.g., 5% means \$5 per \$100 borrowed)
- **Principal/Face value/Par value** - Final repayment amount
- **Maturity date/term to maturity** - When issuer repays principal and payments stop

Valuing a Coupon Bond: Step 1

Bond characteristics:

- Yearly coupon payment: $\$C$
- Number of years: n
- Principal payment: $\$F$
- Interest rate: i

Step 1: Value the bond principal (BP)

$$P_{BP} = \frac{\$F}{(1 + i)^n}$$

Example: \$1000 principal, 6% interest, 30 years:

$$P_{BP} = \frac{\$1000}{(1.06)^{30}} = \$174.11$$

Valuing a Coupon Bond: Step 2

Step 2: Value the coupon payments (CP)

Sum of present values of all coupon payments:

$$P_{CP} = \sum_{k=1}^n \frac{\$C}{(1+i)^k}$$

Simplified formula:

$$P_{CP} = \frac{\$C}{i} \left[1 - \frac{1}{(1+i)^n} \right]$$

Example: \$10 coupon, 6% interest, 2 years:

$$P_{CP} = \frac{\$10}{1.06} + \frac{\$10}{(1.06)^2} = \$9.43 + \$8.90 = \$18.33$$

Valuing a Coupon Bond: Step 3

Step 3: Total coupon bond (CB) value

$$P_{CB} = P_{BP} + P_{CP} = \frac{\$F}{(1+i)^n} + \sum_{k=1}^n \frac{\$C}{(1+i)^k}$$

Key principle: The value of a bond varies inversely with the interest rate

- Higher interest rate → Lower bond price
- Lower interest rate → Higher bond price

Real and Nominal Interest Rates

When borrowing/lending, we care about **goods**, not just dollars!

Nominal interest rate (i_t):

- Expressed in currency units
- Borrow \$1 today $\rightarrow$ Repay $1 + i_t$ dollars next year

Real interest rate (r_t):

- Expressed in baskets of goods
- Borrow 1 basket today $\rightarrow$ Repay $1 + r_t$ baskets next year

Challenge: We observe nominal rates, but real rates matter for decisions.

Deriving the Real Interest Rate

Setup:

- Current price level: P_t (price of one basket of goods)
- Expected price level next year: P_{t+1}^e
- Borrow P_t dollars today $\rightarrow$ Repay $(1 + i_t)P_t$ dollars next year

In terms of baskets:

Repayment next year = $\frac{(1+i_t)P_t}{P_{t+1}^e}$ baskets

Real interest rate formula:

$$1 + r_t = (1 + i_t) \frac{P_t}{P_{t+1}^e}$$

Fisher Relation

Define expected inflation: $\pi_{t+1}^e \equiv \frac{P_{t+1}^e - P_t}{P_t}$

Exact relation:

$$1 + r_t = \frac{1 + i_t}{1 + \pi_{t+1}^e}$$

Fisher relation (approximation when rates are not too large):

$$i_t \approx r_t + \pi_{t+1}^e$$

Key insight: Nominal interest rate is positively related to expected inflation.

Ex-Ante vs. Ex-Post Real Rates

Ex-ante real interest rate ("before the fact"):

- Uses expected inflation: $r_t = i_t - \pi_{t+1}^e$
- Cannot be directly observed (must be estimated)
- Relevant for decision-making

Ex-post real interest rate ("after the fact"):

- Uses actual inflation: $r_t = i_t - \pi_{t+1}$
- Can always be computed after the fact
- Shows what actually happened

Math Appendix

Goal: Simplify $PV = \frac{\$C}{1+i} + \frac{\$C}{(1+i)^2} + \dots + \frac{\$C}{(1+i)^n}$

Step 1: Multiply both sides by $\frac{1}{1+i}$:

$$\frac{PV}{1+i} = \frac{\$C}{(1+i)^2} + \frac{\$C}{(1+i)^3} + \dots + \frac{\$C}{(1+i)^{n+1}}$$

Step 2: Subtract to get:

$$\frac{i}{1+i} \times PV = \frac{\$C}{1+i} - \frac{\$C}{(1+i)^{n+1}}$$

Step 3: Solve for PV:

$$PV = \frac{\$C}{i} \left[1 - \frac{1}{(1+i)^n} \right]$$